CONTACT INFORMATION

Darla Moore School of Business
1014 Greene Street, Columbia, SC,
29208 / USA

E-mail:Zheyu.qi@grad.moore.sc.edu

EDUCATION

University of South Carolina - Darla Moore School of Business. Columbia, SC

- Ph.D. Candidate in Business Administration, Finance Concentration
- Ph.D. Award Expected May 2027
- Advisor: Prof. Allen N. Berger

Clark University - School of Management. Worcester, MA

- Master of Science in Finance, 2021 GPA3.9/4.0
- Area of Specialization: Finance

Inner Mongolia University of Finance and Economics - School of Finance. Hohhot, China

- Bachelor of Economics in Financial Engineering, 2019 GPA3.5/4.0
- Area of Specialization: Financial Engineering

RESEARCH INTEREST

Banking and Financial Institutions, Artificial Intelligence, Household Finance

WORKING PAPERS

Artificial Intelligence, Information Hold-Up, and the Transformation of Relationship Lending (with Allen N. Berger, Raluca A.Roman)
Presentations: EasternFA 2026, University of South Carolina

We provide evidence that bank adoption of artificial intelligence transforms relationship lending while increasing the cost of corporate credit. Using novel data linking AI patents to syndicated loan contracts spanning 1995–2023, we document that AI adoption raises corporate loan spreads by 9 to 13 basis points, with propensity score matching and Bartik-style instrumental variables confirming robustness and suggesting substantially larger effects. The mechanism is converting soft information into proprietary hard information that deepens rather than reduces information asymmetry. The traditional price discount associated with relationship lending erodes substantially at AI-adopting banks, yet relationships persist by migrating to non-price terms – loans become smaller, shorter, and carry fewer covenants –transforming the function of relationships from spread reduction to enhanced rollover discipline. First-wave machine learning (1995-2010) reprices public borrowers; second-wave deep learning and natural language processing (2011-2023) shift burdens toward private borrowers. Borrowers that adopt AI face lower spreads, but this discount disappears at AI-adopting banks, an asymmetry consistent with rent extraction rather than symmetric efficiency gains. Financially stronger banks pass through larger spread increases.

The Algorithmic Banker: AI Adoption and Consumer Service in Retail Banking
(Solo-Authored)

We provide the first large-scale evidence on how artificial intelligence (AI) adoption in retail banking affects consumer outcomes. Using CFPB complaint data (2010–2023), we construct a bank-county-quarter panel linking the universe of consumer complaints to bank-level AI patent filings and show that AI adoption reduces aggregate complaint rates by approximately 30 percent relative to the sample mean. An event-study design confirms parallel pre-trends and reveals that complaint reductions emerge with a two-year lag consistent with gradual technology deployment; propensity score matching and a Bartik-style instrumental variable corroborate a causal interpretation. However, these gains conceal opposing effects across bank services. AI eliminates operational errors in transaction processing and credit reporting while generating new grievances through opaque algorithmic decisions and rigid automation in credit and customer service. AI adoption also improves resolution quality. Adopting banks deliver a higher share of substantive, non-monetary remediation and fewer outright dismissals, indicating that the complaint decline reflects genuine service improvement rather than complaint suppression. Finally, applying dictionary-based textual analysis to complaint narratives, we show that AI adoption reduces discrimination-related complaints. Our findings reveal that the net welfare effect of bank AI adoption is positive in the aggregate but distributionally heterogeneous across the product lines that constitute retail banking, with important implications for the design of consumer financial protection in an era of algorithmic banking.

Artificial Intelligence and the Geography of Credit: Evidence from Global Syndicated Lending (with Omrane Guedhami, Raluca A.Roman)

We examine how bank AI adoption affects cross-border syndicated lending. We find AI-adopting banks expand cross-border lending relative to domestic lending, with effects concentrated among informationally opaque borrowers. AI compresses loan spreads and reduces the geographic distance premium, consistent with algorithmic processing substituting for local knowledge. Exploiting regulatory variation from GDPR (2018), China's PIPL (2021), and the EU AI Act (2024), we identify technology-driven regulatory arbitrage: banks in permissive jurisdictions gain competitive advantages lending to borrowers in restrictive environments. These effects operate through information-processing channels distinct from traditional capital regulatory arbitrage.

Lending in the Age of Cyber Threats: How do lenders Price Loans? (Solo-Authored)
-AFS 2025 Conference - Outstanding Doctoral Research Paper Award
Presentations: 51st EBES Conference Annual Meeting 2025, AFS Annual Conference 2025, University of South Carolina

We examine how lenders price the emerging risks of data breaches in the syndicated loan market. We find that bank lenders increase loan spreads for firms that experienced a data breach during the loan underwriting process. This increase is particularly pronounced for credit lines, rather than term loans. These findings support the Bank Specialness Hypothesis, suggesting that banks maintain a significant informational advantage in pricing new risk factors compared to non-bank lenders. Our results also indicate that firms experiencing a data breach can substantially mitigate these pricing effects through two channels: strong borrower-lender relationships and proactive cybersecurity risk management. This finding complements existing relationship lending studies by incorporating cybersecurity risk.

The Paycheck Protection Program (PPP) from the Small Business Perspective: Did the PPP Help Alleviate Financial and Economic Constraints? (with Allen N. Berger, Jonathan A. Scott, Paul Freed, Siwen Zhang, John. Ampong)
Presentations: 2022 SFA, 2022 FMA, Temple University, University of South Carolina

We employ two novel datasets to evaluate whether the Paycheck Protection Program (PPP) met short-term and longer-term goals. One dataset matches PPP with small business members of the National Federation of Business (NFIB) that report their financial constraints and their employees' economic constraints. We find more alleviation of both constraints for PPP recipients, consistent with short-term goals. We employ data on post- versus pre-crisis county employment, wages, and new businesses to test the longer-term goal of improved community crisis recoveries from the short-term constraint relief. We find to the contrary, worse longer-term recoveries in counties with more short-term constraint relief.

WORK IN PROGRESS

Bank AI adoption and its impact on Mortgage Market (with Allen N. Berger and Raluca A. Roman)
- *Preliminary results documented*

This project examines how bank adoption of artificial intelligence (AI) affects mortgage pricing in the U.S. residential mortgage market. While prior research shows that AI can improve screening accuracy and risk assessment, it remains unclear whether these efficiency gains are passed through to borrowers or instead strengthen lenders' pricing power. We construct a novel bank–loan level dataset linking measures of bank AI adoption to detailed mortgage origination data, including interest rates, borrower characteristics, and loan terms. Our empirical strategy exploits within-bank variation over time, controlling for borrower risk, local housing market conditions, and bank fixed effects. We test two competing hypotheses: an efficiency channel predicting lower mortgage rates due to improved underwriting precision, and a market-power channel predicting higher rates if AI enhances proprietary information and reduces borrower switching. By distinguishing effects across borrower risk segments and competitive environments, the study provides new evidence on the distributional and welfare implications of AI adoption in household credit markets.

Artificial Intelligence and Bank Systemic Risk (with Allen N. Berger and Raluca A. Roman)
- *Preliminary results documented*

This project investigates whether bank adoption of artificial intelligence (AI) stabilizes or amplifies systemic risk in the financial system. AI can enhance credit screening, fraud detection, and portfolio risk management, potentially reducing tail risk at individual institutions. However, AI may also increase systemic fragility by promoting model homogeneity, synchronized trading and lending decisions, procyclical credit adjustments, and greater reliance on opaque algorithmic signals. The project provides new evidence on whether AI improves institutional resilience or instead increases the vulnerability of the banking system to aggregate shocks, with important implications for financial regulation and macroprudential policy.

TEACHING EXPERIENCE

Instructor FINA 365 Corporate Finance (Undergraduate) (In-Person)
Evaluations: 4.0/5.0

Selected Students Comments:

-Professor really cares and takes time to talk out theories, not only showing on slide shows. I think she believes that if you understand it you can talk about it and that leads us to be able to understand more since she makes us talk about what we learn very effortlessly.

-She actually cares about her students and wants them to learn. She makes sure everyone is on pace during class and slows down and helps students if needed.

-She is very down-to-earth and knows how to create a good learning environment.

-The lecture structure resonating with me well. Being a smaller class, we were encouraged to participate more in class which helps me hold the information longer.

Teaching Assistant
2024 Spring FINA 469 Investment Analysis and Portfolio Management (Undergraduate)
2023 Spring FINA 365 Corporate Financial Analysis (Undergraduate)

ACADEMIC SERVICES

Conference Discussions

EasternFA discussant	2026
FMA discussant (x2)	2025
AFS discussant	2025

Internal Services

Exam Coordinator, Department of Finance	2024-2025

HONORS AND AWARDS

Outstanding Doctoral Research Paper Award. AFS 2025 Conference
University of South Carolina Graduate School Travel Grant (x2)
Special Scholarship. Darla Moore School of Business 2025
Second Year Paper Reward. 2024 First Year Paper Reward. 2023
Doctoral Fellowship. Darla Moore School of Business

SKILLS

STATA, Python, Machine Learning, SAS, R, LaTeX, Bloomberg Terminal

REFERENCES

Allen N. Berger
Carolina Distinguished Professor – Montague Osteen Jr. Professor of Banking and Finance
University of South Carolina, Darla Moore School of Business
E-mail: aberger@moore.sc.edu

Eric A. Powers
Chair, Department of Finance – Associate Professor of Finance University of South Carolina, Darla Moore School of Business E-mail: epowers@moore.sc.edu

Raluca A. Roman
Senior Economist
Federal Reserve Bank of Philadelphia E-mail: raluca.roman@phil.frb.org

Omrane Guedhami
C. Russell Hill Professor – professor of international finance Coordinating director of the Ph.D. program in international finance University of South Carolina, Darla Moore School of Business
E-mail: omrane.guedhami@moore.sc.edu